

Part 1: The work

This part of the Statement of Work (SOW) defines the project-specific scope and technical requirements.

1.0 SCOPE

This Statement of Work (SOW) defines the requirements to furnish all labor, materials, and equipment necessary for the complete removal and disposal of the existing chiller and to provide and install a new, fully operational chiller system at Building 149, Pease Air National Guard Base (ANGB).

2.0 BACKGROUND

The 157th Civil Engineer Squadron requires a full replacement of the chiller for Building 149 at Pease ANGB. The building's climate control relies on an existing air-cooled chiller that provides chilled water to air handlers for cooling. This aging system has experienced multiple failures, causing shutdowns that impact mission-critical operations within the facility. Replacement of the chiller is necessary to ensure reliable environmental control and support ongoing missions.

2.1 Period of Performance (PoP): Work shall be completed within 135 calendar days of contract award.

3.0 EXISTING SITE CONDITIONS

Building 149 at Pease ANGB is currently served by an aging air-cooled chiller system that provides chilled water to air handlers for facility-wide cooling. The existing chiller and its associated components are located outside the building and are nearing the end of their service life. This system has become unreliable, experiencing multiple failures that have previously impacted building operations. The existing chiller is not on the base's controls system. The existing base wide control system is integrated with the base-wide Schneider Electric platform, and the new installation must be fully compatible with this infrastructure. All work will take place around the existing chiller footprint, and the contractor will be responsible for verifying all site-specific utility connections and dimensions prior to commencing work.

3.1 Site Operation hours & Constraints

- Normal project work hours are defined as 0600-1530, Monday through Friday.
- Work outside of these hours or on Federal holidays requires written approval from the COR submitted at least 72 hours in advance.

- The site will be closed, and no work is permitted, on all Federal holidays or Drill weekend.

3.2 Base Access & Badging

The contractor will be required to submit all necessary documentation for base access and badging within 10 days of contract award. Additional information will be required to specific site access.

3.3 Quality Control (QC): The contractor shall develop and maintain an effective QC Plan (QCP) to ensure services are performed in accordance with this Statement of Work. The contractor shall develop and implement procedures to identify, prevent, and ensure nonrecurrence of defective performance. The contractor's QCP is the means by which it assures itself that its work complies with the requirements of the contract. At a minimum, the contractor shall develop QC procedures that address the areas identified in Technical Exhibit I, Performance Requirements Summary (PRS). A final QCP shall be submitted to the Contracting Officer (KO) / Contracting Officer Representative (COR) NLT 10 days after contract award. After acceptance of the QCP, the contractor shall obtain the Contracting Officer's (KO's)/COR's acceptance in writing of any proposed changes to its QCP.

3.4 Access and General Protection/Security Policy and Procedures. The contractor shall comply with all applicable installation/facility access and local security policies and procedures, which may be obtained from the Contracting Officer (KO)/ Contracting Officer Representative (COR). The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by the local installation's Security Forces, Director of Emergency Services or local Security Office. The contractor shall ensure compliance with all personal identity verification requirements as directed by DoD, Headquarters Air Force (HAF) and/or local policy. Should the Force Protection Condition (FPCON) change, the Government may require changes in contractor security matters or processes.

1. The contractor shall obtain base identification and vehicle passes, if required, for all contractor personnel who make frequent visits to or perform work on the Air Force installation(s) cited in the contract. Contractor personnel are required to wear or prominently display installation identification badges or contractor-furnished, contractor identification badges while visiting or performing work on the installation.
2. The contractor shall submit a written request on company letterhead to the contracting officer listing the following: contract number, location of work site, start and stop dates, and names of employees and subcontractor employees needing access to the base. The letter will also specify the individual(s) authorized to sign for a request for base identification credentials or vehicle passes. The contracting officer will endorse the request and forward it to the issuing base pass and registration office or Security Forces

for processing. When reporting to the registration office, the authorized contractor individual(s) should provide a valid driver's license, current vehicle registration, valid vehicle insurance certificate, and [insert any additional requirements to comply with local security procedures] to obtain a vehicle pass.

3. During performance of the contract, the contractor shall be responsible for obtaining required identification for newly assigned personnel and for prompt return of credentials and vehicle passes for any employee who no longer requires access to the work site.
4. When work under this contract requires unescorted entry to controlled or restricted areas, the contractor shall comply with AFI 31-101, Integrated Defense, and DODMAN5200.02_AFMAN 16-1405, Air Force Personnel Security Program.
5. Upon completion or termination of the contract or expiration of the identification passes, the prime contractor shall ensure that all base identification passes issued to employees and subcontractor employees are returned to the issuing office.
6. Failure to comply with these requirements may result in withholding of final payment.

3.5 Antiterrorism Awareness Level I Training (AT Level I). All contractor employees, to include subcontractor employees, requiring access to US Government installations, facilities and controlled access areas shall complete AT Level I training within 15 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. AT Level I may be accomplished by a Level I qualified instructor; completing the Force Protection computer-based training (CBT) course on the Advanced Distributed Learning System (ADLS) or Joint Knowledge Online at <http://jko.jten.mil>. Tracking for each contractor or subcontractor employee is the responsibility of the KO or unit AT Representative. The contractor shall submit certificates of completion for each affected contractor and subcontractor employee to the KO within 15 calendar days after completion of training by each employee or subcontractor personnel.

3.6 Communications Security/Information Technology (COMSEC/IT) Security. All communications with DoD organizations are subject to COMSEC review. All telephone communications networks are continually subject to intercept by unfriendly intelligence organizations. DoD has authorized the military departments to conduct COMSEC monitoring and recording of telephone calls originating from, or terminating at, DoD organizations. Therefore, the contractor is advised that any time contractor personnel place or receive a call they are subject to COMSEC procedures. The contractor shall ensure wide and frequent dissemination of the above information to all employees dealing with DoD information. The contractor shall abide by all Government regulations concerning the authorized use of the Government's computer network, including the

restriction against using the network to recruit Government personnel or advertise job openings.

4.0 PROJECT OVERVIEW

The objective of this project is to execute a full, turnkey replacement of the existing air-cooled chiller system at Building 149, Pease Air National Guard Base. The contractor shall be responsible for providing all necessary personnel, supervision, equipment, and materials to complete the effort. This scope includes the systematic removal and proper disposal of the existing chiller and its associated components, followed by the installation, integration, and commissioning of a new, fully operational chiller system. The final deliverable shall be a safe, reliable, and code-compliant cooling system that meets all specifications outlined in this Statement of Work.

4.1 Special Qualifications: Contractors must ensure that any personnel performing under a contract are fully trained, licensed, certified, and otherwise qualified to perform the desired services of similar size and complexity.

4.2 Post Award Conference/Periodic Progress Meetings: The contractor agrees to attend any post award conference convened by the KO in accordance with FAR 42.5. The KO and other Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings, the KO will apprise the contractor of how the Government views the contractor's performance and the contractor shall apprise the Government of problems, if any, being experienced. The contractor shall resolve outstanding issues raised by the Government. Contractor attendance at these meetings shall be at no additional cost to the Government.

4.3 Contract Manager (CM): The contractor shall designate a CM who shall ensure performance under this contract. The name of this person, and an alternate who shall act for the contractor when the CM is absent, shall be designated in writing to the KO. The CM or alternate shall have full authority to act for the contractor on all contract matters relating to daily operation of this contract. The CM shall work through the KO to resolve issues, receive technical instructions, and ensure adequate performance of services. The CM shall ensure that contractor employees do not perform any services outside the scope of the contract without an official modification issued by the KO. The CM shall ensure contractor employees understand that services performed outside the scope of the contract are performed wholly at the expense of the contractor.

4.4 Identification of Contractor Employees: All contractor personnel attending meetings, answering Government telephones and working in other situations where their contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression that they are Government employees. The contractor shall ensure that all documents or reports produced by contractor personnel are suitably marked as contractor products or that contractor participation is appropriately disclosed. The contractor's status as a "contractor" shall be predominantly displayed in all correspondence types (to include signature blocks on email) and dealings with Government or non-Government entities. Contractor personnel shall wear identification badges distinguishing themselves as such. The badges shall have the company name, employee name and the word "contractor" displayed.

4.5 The contractor shall retrieve all identification media (including vehicle passes) from its employees who depart employment for any reason. The contractor shall return all identification media (i.e., badges and vehicles passes) to the KO/COR within 14 days of an employee's departure.

4.6 Combating Trafficking in Persons: The United States Government has adopted a zero tolerance policy regarding trafficking in persons. Contractors and contractor employees shall not engage in severe forms of trafficking in persons during the period of performance of the contract; procure commercial sex acts during the period of performance of the contract; or use forced labor in the performance of the contract. The Contractor shall notify its employees of the United States Government's zero tolerance policy, the actions that will be taken against employees for violations of this policy. The contractor shall take appropriate action, up to and including termination, against employees or subcontractors that violate the US Government policy as described at FAR 22.17.

4.7 Organizational Conflicts of Interest (OCI): The contractor and subcontractor personnel performing services under this contract may receive, have access to or participate in the development of proprietary or source selection information (e.g., cost or pricing information, budget information or analyses, specifications or work statements, etc.) or perform evaluation services which may create a current or subsequent OCIs, as defined in FAR Subpart 9.5. The contractor shall notify the KO immediately whenever it becomes aware that such access or participation may result in any actual or potential OCI and shall promptly submit a plan to the KO to avoid or mitigate any such OCI. The contractor's mitigation plan will be determined to be acceptable solely at the discretion of the KO. In the event the KO unilaterally determines that any such OCI cannot be satisfactorily avoided or mitigated, the KO may impose other remedies as he or she deems necessary, including prohibiting the contractor from participation in subsequent contracted requirements which may be affected by the OCI.

4.8 Utilities: All utilities in the facility will be available for the contractor's use in the performance of this contract. The contractor shall instruct employees in utilities conservation practices. The contractor shall operate under conditions that preclude the waste of utilities, which include turning off the water faucets or valves after using the required amount.

4.9 Government Furnished Property, Material, Equipment and Services (GFP/M/E/S): The Government will provide the property, material, equipment, and/or services listed below solely for the purpose of performance under this contract: NONE

4.9.1 The Government will furnish the necessary workspace for the contractor to perform services outlined in this PWS to include a clear workspace surrounding the site.

5.0 DETAILED DESCRIPTION OF WORK

This part outlines the mandatory requirements for the equipment replacement and its full integration. It specifies that the contractor must furnish all personnel, equipment, tools, materials, supervision, and quality control necessary to replace the failing chilled water system equipment at Bldg 149. The project includes the removal of the existing chiller and buffer tank and the installation of a new, like-for-like, 25-ton modular air-cooled chiller and a 460-gallon buffer tank. The scope covers planning, demolition, installation, and control system integration to provide a complete and usable system.

5.1 Provide a work plan to be approved by the COR prior to performing any work. The work plan will include, at a minimum, a project schedule, a list of all major components and materials by make and model for submittal, and a written description of the execution plan. The schedule must identify the sequence of events from material acquisition through project closeout, including all major events, work phases, equipment deliveries, and outage times. All outages must be submitted to the BCE in writing for approval.

5.2 The chiller shall be commissioned prior to turnover.

5.3 Provide new as-built drawings, schematics, manuals, and specifications in an acceptable format.

5.4 Scheduling of Work: The project schedule shall identify all outage times and durations. All system outages must be submitted to BCE in writing for approval.

5.5 Dismantling: Remove and dispose of the failing chiller in accordance with Federal and New Hampshire Department of Environmental Services regulations. All chiller and associated appurtenances shall be removed. No equipment shall be abandoned in place. Provide COR weight slips for all debris removed from the installation.

5.6 Remove and dispose of the existing chilled water buffer tank and all piping to and from the chiller. Existing piping from the buffer tank to the building is to remain.

5.7 Installation: New chiller shall be placed on the existing concrete pad. The contractor is responsible for modifying the existing pad as needed for the new chiller. Any modifications to the pad shall be of similar construction standards as the original.

5.8 Install new 4" flanged chilled water connections for the new buffer tank.

5.9 Provide and install a new vertical chilled water buffer tank. The tank must be ASME rated for MAWP 125 PSIG @ 180°F, have a 460-gallon capacity, be of steel construction with angle legs, and feature two 4" flanged chilled water connections.

5.10 Provide new electrical wiring, disconnects, circuit breakers, and other electrical appurtenances as required by current UFC and NFPA standards for a fully functional and code-compliant system.

5.11 Provide a 30% glycol solution in the chilled water loop after the new chiller is online.

5.12 Provide and replace all HVAC controllers in the facility with Bacnet compliant, Schneider Electric IP controllers. Perform controller programming and commissioning. Add new graphics to the existing Schneider Electric central Building Management server located in the Civil Engineer facility, B151. The new chiller controls shall all have the same functionality as the existing chiller controls. The new graphics for the new controllers related to the other system components, i.e. boilers, air handlers, valves, supply air, return air, etc., shall have the same points and control functions as the existing controls system. The control points listed in the table below shall be provided at a minimum. No part of the existing controlled HVAC system shall be transferred to the new controls server with associated graphics.

FUNCTION	NAME	DESCRIPTION
PROOFS & SAFTIES	CHLR-S	CHILLER STATUS
	CHLR-SS	CHILLER START-STOP (ON/OFF) COMMAND
	CHLR-MOD	CHILLER MODE - OPERATIONAL CODE
	CHLR-ALM	CHILLER ALARM/FAULT
	CHLR-LOW-T	CHILLER LOW TEMP CUTOUT
	CHLR-LO	SYSTEM 1 LOCKOUT
	P-1 ENABLE	VFD ENABLE

	P-1 SIGNAL	VFD SIGNAL
	P-1 ALARM	VFD ALARM
	P-2 ENABLE	VFD ENABLE
	P-2 SIGNAL	VFD SIGNAL
	P-2ALARM	VFD ALARM
COMPRESSOR	COMP1-S	COMPRESSOR 1 RUN STATUS
	COMP2-S	COMPRESSOR 2 RUN STATUS
EVAPORATOR	EVAP-PMP-S	EVAPORATOR PUMP STATUS (FLOW PROOF)
CONDENSOR	COND-FANS-S	CONDENSER FAN STATUS
SYSTEM	LVG-WATER-T	CHILLED WATER TEMP LEAVING CHILLER
	RET-WATER-T	RETURN CHILLED WATER TEMPERATURE
	CONTMODE	CONTROL MODE - LOCAL OR REMOTE
	CWS-T-SP	CHILLED WATER SUPPLY TEMP SETPOINT
	CHLR-RUN-H	CHILLER RUNTIME HOURS
OTHER POINTS	CHLR-CL	CURRENT LIMIT
	CHLR-SUC-T	SUCTION TEMPERATURE
	CHLR-DIS-T	DISCHARGE TEMPERATURE
	CHLR-OIL-T	OIL TEMPERATURE
	CHLR-OIL-P	OIL PRESSURE
	CHLR-SUC-P	SUCTION PRESSURE
	CHLR-DIS-P	DISCHARGE PRESSURE
	CHLR-FL-AMP	FULL LOAD AMP
	CHLR-SUCSH-T	SUCTION SUPERHEAT TEMPERATURE
	CHLR-DISSH-T	DISCHARGE SUPERHEAT TEMPERATURE
	OA-T	OUTSIDE AIR AMBIENT TEMPERATURE

5.13 Provide a new modular 25-ton chiller, consisting of a 10-ton and a 15-ton module. The new chiller shall have the same electrical characteristics as the failed chiller. The unit shall be provided with internal pumps (with VFDs) providing the same flow/pressure as the existing, an internal expansion tank of the same capacity, and an internal glycol feed. Make all line, load, controls, and any other connections to provide a full, safe, and properly functioning system.

5.14 Provide new pressure gauges, strainers with blow-down, pressure taps, isolation valves, unions, vibration isolation, check valves, flow meters, and all associated appurtenances as required per manufacturer's instructions for a fully functional system.

5.15 Project Closeout and Deliverables: The contractor will provide all equipment manuals & specifications and manufacturer warranty information. The contractor will also provide an orientation on the operation and maintenance of the new equipment to Pease ANGB maintenance personnel.

5.16 The contractor shall perform all testing, commissioning, and acceptance activities. A detailed Test and Commissioning Plan must be submitted to the COR for approval prior to execution. To commission the new system, the contractor shall run the unit under full capacity for no less than 72 hours at a temperature to be provided by the Government. The contractor will demonstrate the system's full operational capabilities, including performance under various load conditions, to be witnessed by the COR or a designated government representative. A final commissioning report, including all test results, shall be provided as part of the project closeout documentation prior to acceptance.

Paragraph	Deliverable	Required by Government
3.3	Quality Control Plan (QCP)	NLT 10 days after contract award.
3.4	Base Access and Personnel Information	Prior to any work commencing on the installation.
3.5	AT Level I Training Certificates	Within 15 days of training completion (Training due within 15 days of contract start).
5.1	Work Plan and Schedule	Prior to purchasing any materials.
5.3 & 5.15	As-Built Drawings, Schematics, Manuals, & Warranty Info	Upon project completion.
5.15	Personnel Training and Orientation	Upon project completion.

5.16	Test and Commissioning Plan	Prior to execution of testing activities.
5.16	Final Commissioning Report	As part of project closeout documentation, prior to acceptance.